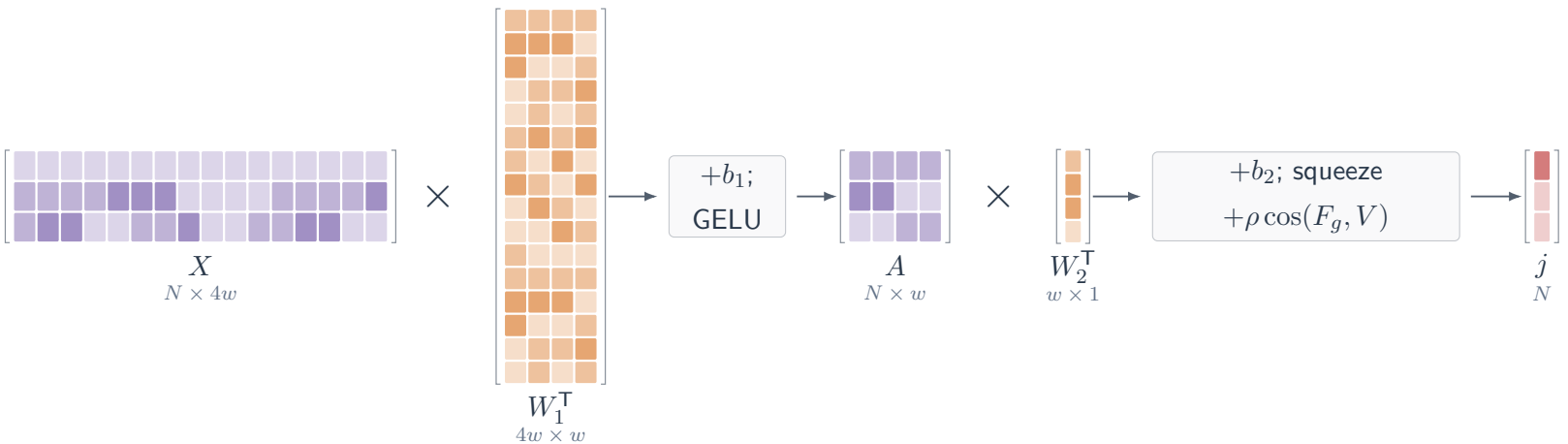
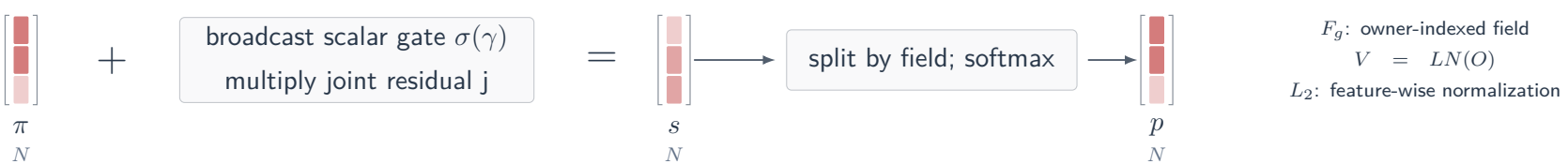
$$X = [F_g, V, F_g \odot V, |F_g - V|], \quad j = \rho \cos(F_g, V) + \mathrm{MLP}(X), \quad s = \pi + \sigma(\gamma) j$$

Joint residual: 4096-to-1024-to-1 MLP, plus field-option cosine



Lexical prior plus gated residual, then field-local normalization

$$\pi_n = \rho_p \langle L_2(L_n), L_2(U_{owner(n)} + g) \rangle, \quad g = (LN_d(H))_{-1}$$



Axes $d = 5120,\ w = 1024$; N total options; w=1024. Four width-w feature parts concatenate to width 4w; both 4w contraction edges match. Bias terms are included in the MLP.

Objects Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes.

Mechanism Two option-to-memory cross-attention layers, then four field decoder layers. Fields communicate globally; lexical prior plus a gated cosine/MLP term gives scores.

Released source exposes inference and head configuration, not the training loss or historical optimization. Model-card base name and config class are recorded separately.